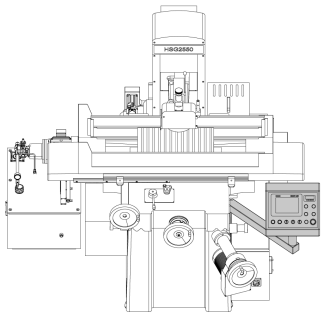
# 5.1 HSG - OPERATION

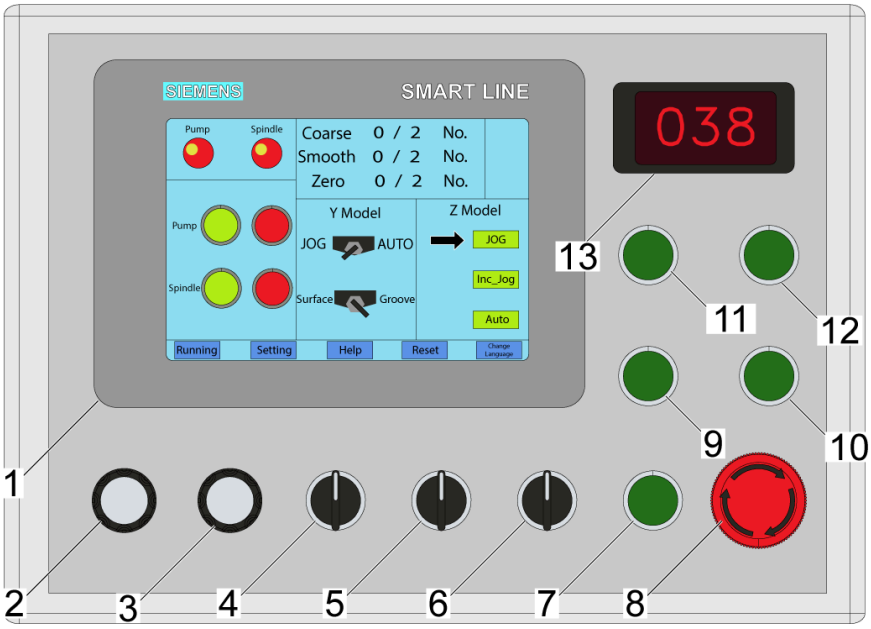
## **Haas Surface Grinder (HSG)** - Operation

## Overview



This chapter will provide details on how to operate the surface grinder. It will cover how to navigate the main control as well as other basic operation details.

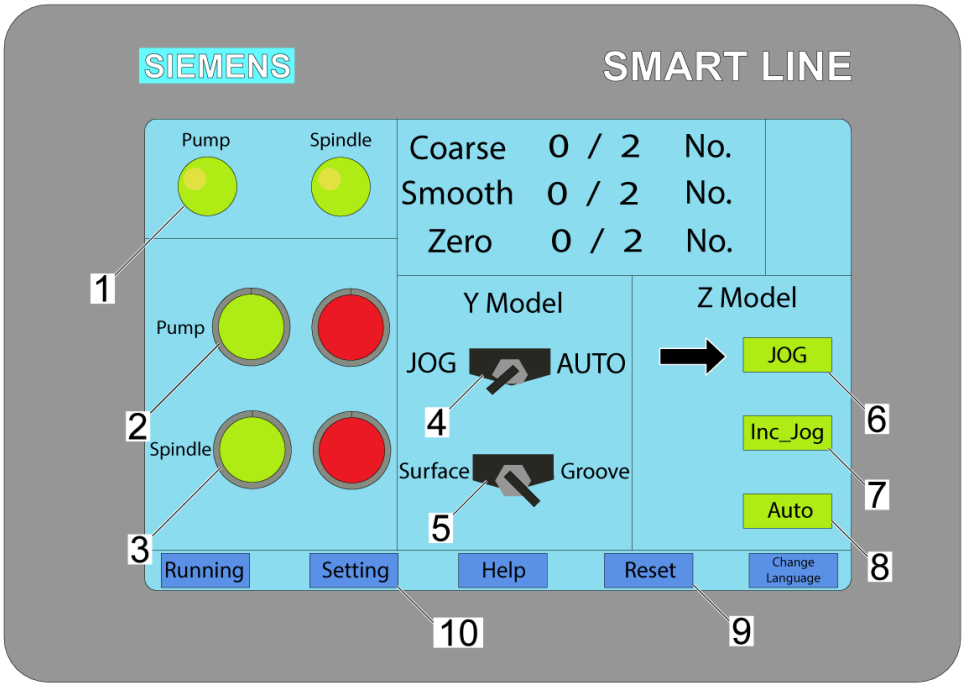
## Main Control




| NUMBER | TYPE | NAME |
|---|---|---|
| 1 | Screen | Main screen |
| 2 | Dials | Electro-magnetic chuck voltage |
| 3 | | Y-axis intermittent feed |
| 4 | Switches | Work lamp on/off |
| 5 | | Coolant pump on/off |
| 6 | | Electro-magnetic chuck on/off |

| | | |
|---|---|---|
| 7 | Buttons | Power up |
| 8 | | E-Stop |
| 9 | | Z-axis rapid up |
| 10 | | Z-axis rapid down |
| 11 | | Y-axis rapid forward |
| 12 | | Y-axis rapid backwards |
| 13 | Display | Magnetic chuck display |

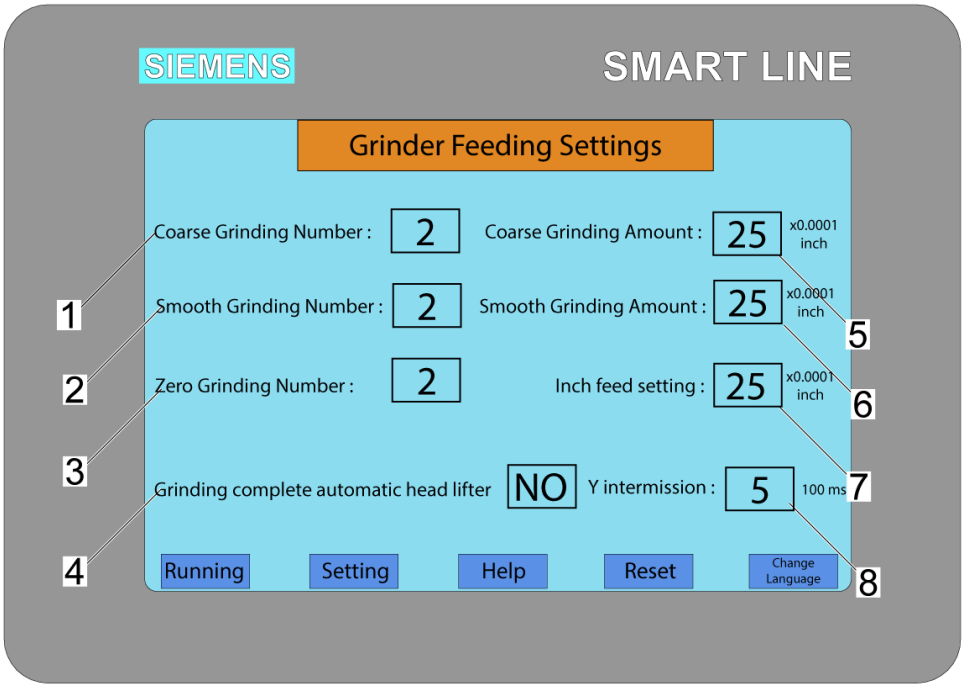
## Running Page




| NUMBER | TYPE | NAME |
|---|---|---|

| | | |
|---|---|---|
| 1 | Screen | Pump/Spindle On/Off indicator lights |
| 2 | | Pump On/Off buttons |
| 3 | | Spindle On/Off buttons |
| 4 | | Y-Axis mode selection (Jog or Auto) |
| 5 | | Y-Axis mode selction (Surface / Groove) |
| 6 | | Z-Axis Jog mode buton |
| 7 | | Z-Axis Incremental Jog mode button (Controlled usig inch feed setting) |
| 8 | | Z-Axis Auto mode button (Controlled using settings page) |
| 9 | | Reset button (used after automatic mode has completed its cycle) |
| 10 | | Setting button |

## Settings Page



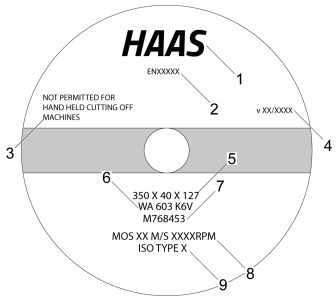

| NUMBER | TYPE | NAME |
|---|---|---|

| | | |
|---|---|---|
| 1 | Screen | Number of Coarse grinding passes |
| 2 | | Number of Smooth grinding passes |
| 3 | | Number of zero (no grinding) passes |
| 4 | | Automatic head lifter On/Off |
| 5 | | Coarse grinding cutting amount |
| 6 | | Smooth grinding cutting amount |
| 7 | | Z-axis incremental job amount |
| 8 | | Y-axis movement time |

# 5.2 HSG - GRINDING WHEEL IDENTIFICATION

## **Haas Surface Grinder (HSG)** - Grinding Wheel Identification

### Grinding wheel Identification




1. Trade Mark
2. Test Record
3. Restriction of Use
4. Expiration Date
5. Dimensions in mm
6. Specification Mark
7. Code Number
8. Maximum Operating Speed
9. Iso Type No (Shape)

### Grinding Wheel Abrasives

| ABRASIVE | ABRASIVE DENOTATION |
|---|---|
| Brown Aluminum Oxide | A |
| White Aluminum Oxide | WA |
| Mixture of brown and white Aluminum Oxide | DA or MA |
| Pink Aluminum Oxide (dark) | PA |
| Pink Aluminum Oxide (light) | PWA |
| Black Silicone Carbide | C |
| Green Silicone Carbide | GC |
| Combination of black and green Silicone Carbide | CGC |
| Blend of Aluminum Oxide and Silicone Carbide | AC |

| | |
|---|---|
| Zirconia with Brown Aluminum Oxide | ZA |
| Micro Crystalline Grains | MCA |
| Ruby Grains | RB |
| Semi friable Aluminum Oxide | FS |
| Sol Gel Grain | CG |

# 5.3 HSG - GRINDING WHEEL REPLACEMENT AND BALANCING

## **Haas Surface Grinder (HSG)** - Grinding Wheel Replacement and Balancing

### Replacement

1
Remove the grinding wheel from the machine using the provided wheel removal tool.

Screw on the removal bolt in the clockwise direction until the grinding wheel is off the taper.

**Note :** The nut thread is left handed so ensure that the wrench is rotated clockwise.

2
Clean all mating surfaces and place the the wheel balancing base and height gauge on top of a level granite table.

Sweep an indicator across the high point of the balancing rail and adjust the bottom 2 knobs untl all 4 corners read withing 0.0001" (0.0025mm).

3
Place the tapered arbor firmly into the grind wheel hub. Place the arbor onto the balancing rails and monitor the rotation of the wheel then let it come to rest.

Mark the heavy and light points on the wheel using chalk and add a balance weight to the light side of the wheel. Equally space and place the other 2 balance weight around the wheel hub.

Lift and rotate the wheel 90 degrees and rest onto the balancing rails to observe movement. Adjust the location of the weight accordingly until the wheel is able to be placed on all 4 quadrants without movement,

4

Place wheel hub through rear bore of the grind wheel then place the hub disk on the opposing side and tighten the hub nut.

Clean the spindle taper then place the wheel firmly on it and tighten the wheel nut in the anti-clockwise direction.

5

Use the handle to manually move the diamond pen on top of the grind wheel.

Rotate the adjustment knob until the pen touches the grinding wheel then return the ram off of the wheel.

Adjust the knob then pull the handle and feed the diamond per fully accross the grind wheel.

Repeat until the grind wheel is fully dressed.